

IKERASSUAQ PRINS CHRISTIAN, Greenland

WMO: 043900

Lat: 60.055N

Lon: 43.165W

Elev: 62

StdP: 14.66

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.1	15.3	-10.8	3.1	18.0	-7.1	3.8	18.4	66.7	30.4	63.1	29.3	24.1	270	0.969

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	57.6	47.6	54.6	46.2	52.2	44.9	49.8	55.1	48.0	52.0	46.5	50.1	13.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
47.3	48.3	49.8	45.7	45.5	48.3	44.3	43.1	47.3	20.2	55.6	19.2	52.3	18.5	50.1	59.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.1	47.4	41.1	10.2	63.8	4.4	3.9	7.0	66.6	4.4	68.8	1.9	71.0	-1.3	73.8
			7.9	52.8	3.8	3.2	5.1	55.1	2.9	57.0	0.8	58.8	-1.9	61.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	26.2	25.5	26.2	31.8	36.5	41.0	44.9	45.8	42.0	36.2	31.6	28.2
	DBStd	8.40	5.02	5.42	5.32	3.94	3.52	3.98	4.04	4.07	3.92	4.12	4.04	4.82
	HDD50	5602	738	687	737	545	419	272	165	143	242	427	552	675
	HDD65	11053	1203	1107	1202	995	884	720	622	596	690	892	1002	1140
	CDD50	24	0	0	0	0	0	2	8	12	2	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.3	19.5	20.6	19.1	16.8	14.8	12.6	10.8	10.0	12.8	14.1	15.5	17.3
Precipitation	PrecAvg	82.6	8.7	6.7	5.4	6.1	5.2	5.8	5.0	6.9	8.4	6.6	7.0	8.2
	PrecMax	129.5	20.2	19.4	12.4	16.6	13.3	18.0	12.7	21.6	19.9	27.9	21.5	26.5
	PrecMin	43.2	2.2	0.2	0.2	0.9	0.6	0.3	1.0	0.7	1.8	0.9	2.1	2.2
	PrecStd	26.4	4.3	4.8	3.2	3.9	3.2	3.6	3.0	4.2	4.3	5.4	4.6	5.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	37.6	40.2	46.9	52.6	57.6	62.9	63.7	56.7	51.3	45.1	46.3
		MCWB	36.3	36.2	37.0	40.6	42.2	47.7	51.7	51.2	44.9	40.6	38.9	38.1
	2%	DB	36.3	36.1	37.1	41.7	47.8	53.1	57.7	57.7	52.8	46.3	41.0	39.3
		MCWB	34.2	35.1	34.6	36.3	39.8	44.4	48.3	48.1	43.1	39.2	36.5	36.5
	5%	DB	34.5	34.3	35.0	39.1	44.5	50.0	53.9	54.5	50.3	43.7	38.7	36.7
		MCWB	32.7	32.4	32.8	34.6	38.4	42.9	46.2	46.8	42.8	37.8	34.5	33.9
	10%	DB	32.8	32.3	33.2	37.2	41.6	47.4	51.4	52.1	47.8	41.8	37.0	34.7
		MCWB	31.2	30.3	31.0	33.8	37.4	41.9	45.2	45.7	41.5	36.7	33.1	32.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	36.9	38.9	41.1	44.1	49.6	52.9	53.1	49.1	44.0	41.8	40.0
		MCDB	38.5	37.1	39.4	45.7	49.2	55.2	59.8	60.2	52.0	47.5	42.8	42.9
	2%	WB	34.8	35.5	35.5	37.8	41.2	45.9	49.4	50.3	46.2	41.3	37.8	36.7
		MCDB	35.6	35.9	36.4	40.2	45.1	50.0	55.5	55.2	49.5	44.1	39.6	38.6
	5%	WB	33.1	33.0	33.4	36.0	39.6	44.0	47.6	48.3	44.1	39.5	35.8	34.5
		MCDB	34.0	33.7	34.6	37.8	42.5	47.9	51.8	52.1	47.7	42.1	37.6	36.4
	10%	WB	31.6	31.0	31.7	34.6	38.2	42.8	46.2	46.8	42.9	37.9	34.2	32.7
		MCDB	32.6	31.9	32.9	36.4	40.9	46.2	49.9	50.2	46.2	40.7	36.1	34.1
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	5.9	6.2	6.9	8.4	8.9	7.9	6.8	5.8	5.2	5.2
		MCDBR	6.4	6.5	6.9	9.2	12.1	13.4	13.9	12.7	10.8	8.5	7.2	6.7
		MCWBR	5.8	6.2	5.8	6.1	7.2	7.7	7.4	6.7	6.3	5.7	5.9	5.5
	5% WB	MCDBR	6.4	6.2	6.1	7.9	9.5	11.4	11.6	10.3	8.6	7.2	6.5	6.3
		MCWBR	6.2	6.3	5.7	6.2	6.5	7.3	7.2	6.5	6.1	5.7	6.1	5.7
Clear-Sky Solar Irradiance	taub	0.272	0.289	0.306	0.349	0.352	0.356	0.356	0.363	0.339	0.293	0.250	0.226	
	taud	2.065	2.254	2.293	2.189	2.248	2.313	2.382	2.362	2.364	2.347	2.162	1.980	
	Ebn at Noon	149	227	262	268	276	277	274	262	248	223	163	118	
	Edh at Noon	15	21	28	38	38	37	34	32	27	19	13	10	
All-Sky Solar Radiation	RadAvg	95	292	739	1277	1667	1676	1589	1235	824	443	148	60	
	RadStd	12	36	60	118	181	155	89	103	87	49	16	5	

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+1.57	+5.22	N/A	+1.10	N/A	-1.13	-490	-464	N/A

Nomenclature: See separate page